Status	Finished
Started	Saturday, 24 October 2026, 9:41 PM
Completed	Saturday, 24 October 2026, 9:46 PM
Duration	5 mins 4 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By using a hash function
c. By comparing each element sequentially
d. By dividing the list length by two ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

a. Hash search
b. Binary search
c. Both (1) & (3)
d. Linear search ✓

Question **3**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(log n) ✓
b. O(1)
c. O(n log n)
d. O👎

Question **4**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O👎 ✓
- b. O(1)
- c. O(n log n)
- d. O(log n)

Question **5**

Correct

Mark 1.00 out of 1.00

Given an array arr = {45,77,89,90,94,99,100} and key = 99; what are the mid values(corresponding array elements) in the first and second levels of recursion?

- a. 90 and 94
- b. 89 and 99
- c. 90 and 99 ✓
- d. 89 and 94

Question **6**

Correct

Mark 1.00 out of 1.00

In the context of searching, what is a successful search?

- a. When the element is found in the list ✓
- b. When the list is sorted
- c. When the list contains duplicate elements
- d. When the search algorithm finishes

Question **7**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 1000
- b. 500
- c. 100
- d. 10 ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Inserting elements into a list
b. Sorting elements in a list
c. Deleting elements from a list
d. Determining whether an element is present in a list ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

a. The search continues until the end of the list
b. The search backtracks to find duplicate elements
c. The search starts over from the beginning
d. The search stops immediately ✓

Question **10**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓
b. log n
c. n-1
d. n/2

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. Must use a sorted array
c. There must be a mechanism to access middle element directly
d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **12**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is small and sorted
b. When the list is small and unsorted ✓
c. When the list is large and unsorted
d. When the list is large and sorted

Question **13**

Correct

Mark 1.00 out of 1.00

The average case occurs in the linear search algorithm

a. Item is the last element in the array or item is not there at all
b. When the item is somewhere in the middle of the array ✓
c. When the item is not the array at all
d. When the item is the last element in the array

Question **14**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

a. The list is sorted
b. The search continues from the beginning
c. The search continues in the left or right sublist ✓
d. The search stops

Question **15**

Correct

Mark 1.00 out of 1.00

Which method of searching involves sequentially comparing each element until a match is found?

a. Linear search ✓
b. Hashing
c. Binary search
d. Jump search